

20. Ans: C

Using $I = nevA$, drift speed $v = \frac{I}{neA}$

$$= \frac{I}{\frac{N}{V}eA}$$

$$= \frac{2.0}{\frac{3.4 \times 10^{22}}{4.0 \times 10^{-7}} \times 1.6 \times 10^{-19} \times 3.0 \times 10^{-6}}$$

$$= 4.9 \times 10^{-5} \text{ m s}^{-1}$$